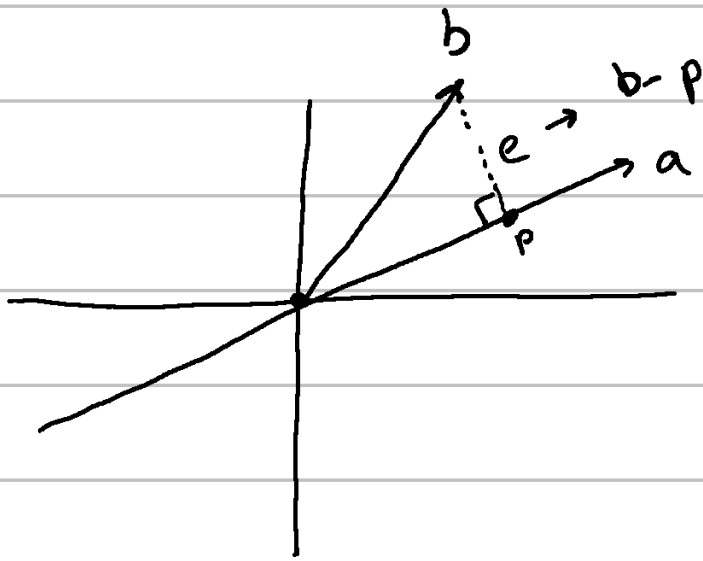


Lecture 15



- # Projection of a vector onto a subspace is its closest distance to it.
- # It is the point that forms an orthogonal vector to subspace to $\vec{b}$.

• $p = x a$ [Projection onto a line]

$$a^T (b - x a) = 0$$

$$a^T b - x a^T a = 0$$

$$x a^T a = a^T b$$

$$x = \frac{a^T b}{a^T a}$$

$$p = ax = a \frac{a^T b}{a^T a}$$

We can say that there is a matrix that acts on b and produces the projection.

$$p = P b$$

$$P = \frac{a a^T}{a^T a}$$

$\rightarrow$ symmetric
 $\rightarrow c(P) = \lambda a$
 $\rightarrow \text{rank} = 1$

$$P^T = P$$

Projection of the projection remains the same. i.e.

$$P^2 = P \quad !!$$

$$P^T = P \quad !$$

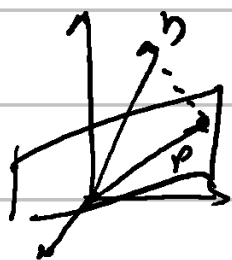
The reason why projections are important $\rightarrow$

- Suppose there is no solⁿ for $Ax = b$

- Then we find the closest projection of b onto the column space of A .

- Solve $A\hat{x} = p$!!!

⇒ Projection onto spaces →



a_1 a_2

basis of plane

$$A = \begin{bmatrix} 1 & 1 \\ a_1 & a_2 \\ 1 & 1 \end{bmatrix} \rightarrow c(A) = \text{plane}$$

$$\boxed{e = b - p}$$

→ perpendicular to plane
↳ perpendicular to both normals

$$A^T(b - p) = 0$$

$$\begin{cases} a_1^T(b - p) = 0 \\ a_2^T(b - p) = 0 \end{cases}$$

$$A^T b - A^T p = 0$$

$$A^T p = A^T b$$

$$p = A\hat{x}$$

$$A^T A \hat{x} = A^T b$$

$$x = (A^T A)^{-1} A^T b$$

P = projection matrix $\rightarrow$

$$P = A (A^T A)^{-1} A^T b$$

$$(A^T A)^{-1} \neq A^{-1} (A^T)^{-1} \text{ because}$$

e is in the nullspace of A^T ! A might not be a square matrix

$$e \in N(A^T)$$

$$e \perp C(A) \quad \underline{\underline{\text{Yes}}}$$

• If A is a square matrix that is invertible then

$$P = A A^{-1} (A^T)^{-1} A^T b = I b$$

That means, col space of full rank square matrix will definitely contain b !!!

$$P = A (A^T A)^{-1} A^T$$

symmetric

$$P^2 = P$$

$$P^T = P$$

$P^2 = P$ proof

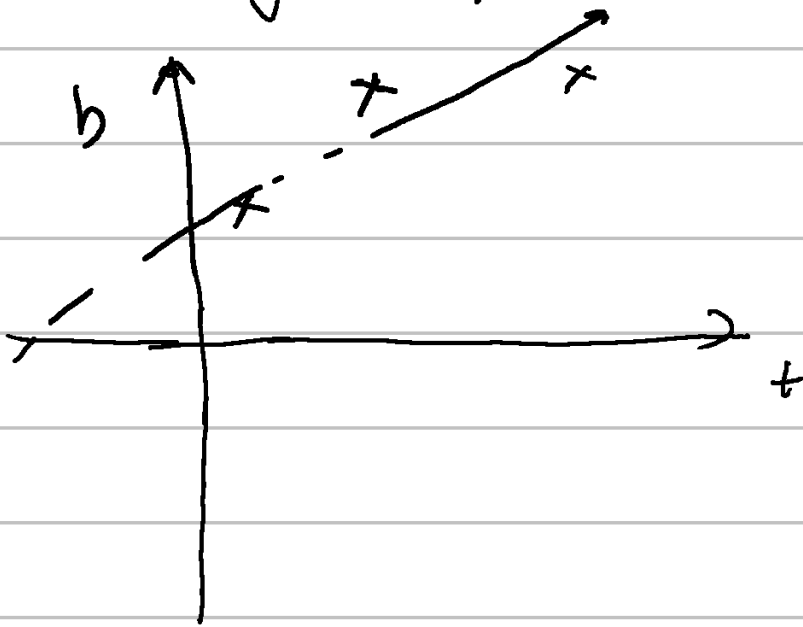
$$(A (A^T A)^{-1} A^T) (A (A^T A)^{-1} A^T)$$

$$\rightarrow A \underbrace{(A^T A)^{-1} (A^T A)}_I (A^T A)^{-1} A^T$$

$$\Rightarrow \underline{\underline{A (A^T A)^{-1} A^T}}$$

Least squares

- Fitting by a line



(1, 1)

(2, 2)

(3, 2)

We have to find the line that best fits these points.

$$\Rightarrow b = c + d t$$

For given points

$$\Rightarrow \left. \begin{array}{l} c + d = 1 \\ c + 2d = 2 \\ c + 3d = 2 \end{array} \right\} \text{unsolvable}$$

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

So in this case we solve

$$A^T A \hat{x} = A^T b !$$